



**Year 11 Mathematics Extension 1**  
**Assessment Task 2**  
**Term 3, Week 4, 2022**

Name: .....

Teacher: .....

**Thursday 11<sup>th</sup> August, 2022**

**Setter: NAI**

**General Instructions**

- **Working time – 45 minutes plus 2 minutes reading time.**
- Answer all questions on the lined paper provided using black or blue pen.
- NESA approved calculators may be used.
- Show all relevant mathematical reasoning and/or calculations.
- **Total marks – 32.**
- Attempt Questions 1–6.

---

| <b>Question 1</b> | <b>(5 marks)</b>                                                                            | <b>Marks</b> |
|-------------------|---------------------------------------------------------------------------------------------|--------------|
| (a)               | (i) Factorise $3a^2 - 10a + 3$ .                                                            | <b>1</b>     |
|                   | (ii) Hence or otherwise solve, $3(\log_3 x)^2 - 10\log_3 x + 3 = 0$ , for $x$ .             | <b>2</b>     |
| (b)               | Differentiate $e^{3x}(x^2 + 1)$ with respect to $x$ , leaving your answer fully factorised. | <b>2</b>     |

**End of Question 1**

| <b>Question 2</b> | <b>(5 marks) (Start a new page)</b>                                                                               | <b>Marks</b> |
|-------------------|-------------------------------------------------------------------------------------------------------------------|--------------|
| (a)               | (i) Find the derivative of $y = e^x - x$ .                                                                        | <b>1</b>     |
|                   | (ii) Hence find the equation of the tangent to the curve at the point where the curve cuts the y-axis.            | <b>2</b>     |
| (b)               | Find the exact value of $\cos\left[\sin^{-1}\left(\frac{5}{13}\right) + \tan^{-1}\left(\frac{3}{5}\right)\right]$ | <b>2</b>     |

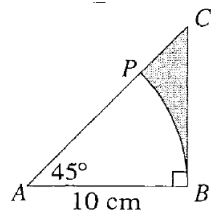
**End of Question 2**

**Question 3** (5 marks) (Start a new page)**Marks**

(a) (i) Prove that  $\frac{\sin 2x}{1 - \cos 2x} = \cot x$ . 2

(ii) Hence find the value for  $\cot 67\frac{1}{2}^\circ$  in simplest surd form. 1

- (b) In the diagram  $\triangle ABC$  is right-angled at  $B$ ,  $AB = 10$  cm and  $\angle A = 45^\circ$ . The circular arc  $BP$  has a centre  $A$  and radius  $AB$ . It meets the hypotenuse  $AC$  at  $P$ .



(i) Find the exact area of sector  $ABP$ . 1

(ii) Hence find the exact area of the shaded portion  $BCP$ . 1

**End of Question 3****Question 4** (5 marks) (Start a new page)**Marks**

Ice-cream is taken out of a freezer at  $-10^\circ\text{C}$  into a room where the air temperature is  $24^\circ\text{C}$ . The rate at which the ice-cream warms follows Newton's law, that is  $\frac{dT}{dt} = -k(T - 24)$ , where  $k$  is a positive integer and  $t$  is measured in minutes and  $T$  in  $^\circ\text{C}$ .

(i) Show that the function  $T = 24 - Ae^{-kt}$ , where  $A$  is a constant, provides this rate of change. 1

(ii) Hence find the value of  $A$ . 1

(iii) The temperature of the ice-cream reaches  $2^\circ\text{C}$  in 1.5 hours. Find the exact value of  $k$ . 2

(iii) Find the temperature of the ice-cream after a further 1.5 hours correct to 2 decimal places. 1

**End of Question 4**

**Question 5** (6 marks) (Start a new page)**Marks**

(a) (i) Prove that:  $\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{x+h}} = \frac{h}{(x+h)\sqrt{x} + x\sqrt{x+h}}$ . **3**

(ii) Hence, find  $\frac{d}{dx} \sqrt{\frac{2}{x}}$  from first principles. **3**

**End of Question 5****Question 6** (6 marks) (Start a new page)**Marks**

(a) Solve  $\sin x + \cos x = -1$ , for  $0 \leq x \leq 2\pi$ , by using the substitution  $t = \tan \frac{x}{2}$ . **2**

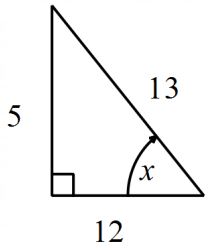
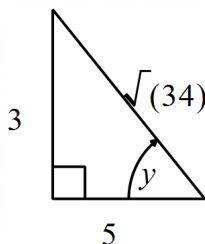
(b) Using  $t$ -formula, prove that:  $\tan 2\theta(\sin 2\theta - \tan \theta) = \sin 2\theta \tan \theta$ . **2**

(c) Using at least a third of a page, sketch the function  $y = \sin(\cos^{-1} x)$ . **2**

**End of Assessment Task**

**Blank page**

## 2022 Year 11 Mathematics Ext1 AT2 Solutions

| Suggested Solution (s)                                                                                                                                                                      | Comments                              | Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                             | Comments                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Question 1</b>                                                                                                                                                                           |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                    |                                           |
| ai) $(3a-1)(a-3)$                                                                                                                                                                           | 1 mark                                | $\cos\left(\sin^{-1}\left(\frac{5}{13}\right) + \tan^{-1}\left(\frac{3}{5}\right)\right)$                                                                                                                                                                                                                                                                                                                          |                                           |
| ii) $(3\log_3 x - 1)(\log_3 x - 3) = 0$<br>$3\log_3 x - 1 = 0$ or $\log_3 x - 3 = 0$<br>$\log_3 x = \frac{1}{3}$ or $\log_3 x = 3$<br>$x = 3^{\frac{1}{3}} = \sqrt[3]{3}$ or $x = 3^3 = 27$ | 1 mark<br><br>1 mark for both answers | $= \cos\left(\sin^{-1}\left(\frac{5}{13}\right)\right) \cos\left(\tan^{-1}\left(\frac{3}{5}\right)\right) - \sin\left(\sin^{-1}\left(\frac{5}{13}\right)\right) \sin\left(\tan^{-1}\left(\frac{3}{5}\right)\right)$<br>$= \frac{12}{13} \times \frac{5}{\sqrt{34}} - \frac{5}{13} \times \frac{3}{\sqrt{34}}$<br>$= \frac{45}{13\sqrt{34}} \times \frac{\sqrt{34}}{\sqrt{34}}$<br>$= \frac{45\sqrt{34}}{442}$      | 1 mark (no penalty for not rationalising) |
| b) let $y = e^{3x}(x^2 + 1)$<br>let $u = e^{3x}$ and $v = x^2 + 1$<br>$u' = 3e^{3x}$ and $v' = 2x$<br>$y' = 3e^{3x}(x^2 + 1) + e^{3x} \times 2x$<br>$y' = e^{3x}(3x^2 + 2x + 3)$            | 1 mark<br><br>1 mark                  | <b>Alternate Solution</b><br>Let $x = \sin^{-1}\left(\frac{5}{13}\right)$ , $\sin x = \frac{5}{13}$<br>$y = \tan^{-1}\left(\frac{3}{5}\right)$ , $\tan y = \frac{3}{5}$<br>$\cos(x + y) = \cos x \cos y - \sin x \sin y$<br>$= \frac{12}{13} \times \frac{5}{\sqrt{34}} - \frac{5}{13} \times \frac{3}{\sqrt{34}}$<br>$= \frac{45}{13\sqrt{34}} \times \frac{\sqrt{34}}{\sqrt{34}}$<br>$= \frac{45\sqrt{34}}{442}$ | Use diagrams as starting point            |
| <b>Question 2</b>                                                                                                                                                                           |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                    |                                           |
| 2ai) $y' = e^x - 1$                                                                                                                                                                         | 1 mark                                |                                                                                                                                                                                                                                                                                                                                                                                                                    |                                           |
| aii) when $x = 0$ , $y = e^0 - 0 = 1$<br>$\therefore$ the curve cuts the $y$ -axis at $(0, 1)$<br>$\therefore$ the equation of the tangent is:<br>$y - 1 = (e^0 - 1)(x - 0)$<br>$y = 1$     | 1 mark<br><br>1 mark                  |                                                                                                                                                                                                                                                                                                                                                                                                                    | 1 mark for the final answer               |
| b)                                                                                                                                                                                          |                                       | <b>Question 3</b>                                                                                                                                                                                                                                                                                                                                                                                                  |                                           |
|                                                                                                           |                                       | 3ai) $LHS = \frac{\sin 2x}{1 - \cos 2x}$<br>$= \frac{2 \sin x \cos x}{1 - (\cos^2 x - \sin^2 x)}$<br>$= \frac{2 \sin x \cos x}{1 - \cos^2 x + \sin^2 x}$<br>$= \frac{2 \sin x \cos x}{\cos^2 x + \sin^2 x - \cos^2 x + \sin^2 x}$<br>$= \frac{2 \sin x \cos x}{2 \sin^2 x}$<br>$= \frac{\cos x}{\sin x}$<br>$= \cot x$                                                                                             | 1 mark                                    |
|                                                                                                           | 1 mark for both the diagrams          |                                                                                                                                                                                                                                                                                                                                                                                                                    | 1 mark                                    |

**2022 Year 11 Mathematics Ext1 AT2 Solutions**

| Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Comments                                                                               | Suggested Solution (s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Comments                                                                                          |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--|
| <p>3aii) <math>\cot 67\frac{1}{2}^\circ = \frac{\sin 2 \times 67\frac{1}{2}^\circ}{1 - \cos 2 \times 67\frac{1}{2}^\circ}</math></p> <p><math>\cot 67\frac{1}{2}^\circ = \frac{\sin 135^\circ}{1 - \cos 135^\circ}</math></p> <p><math>\cot 67\frac{1}{2}^\circ = \frac{1}{\sqrt{2}} \div \left(1 - \left(-\frac{1}{\sqrt{2}}\right)\right)</math></p> <p><math>\cot 67\frac{1}{2}^\circ = \frac{1}{\sqrt{2}} \div \left(\frac{\sqrt{2}+1}{\sqrt{2}}\right)</math></p> <p><math>\cot 67\frac{1}{2}^\circ = \frac{1}{\sqrt{2}+1} = \sqrt{2}-1</math></p> <p>3bi) Area of Sector = <math>\frac{45^\circ}{360^\circ} \times \pi \times 10^2</math></p> <p>Area of Sector = <math>\frac{100\pi}{8}</math></p> <p>Area of Sector = <math>\frac{25\pi}{2}</math></p> <p>bii)</p> <p>Area of shaded portion = <math>\frac{1}{2} \times 10 \times 10 - \frac{25\pi}{2}</math></p> <p>Area of shaded portion = <math>\frac{100 - 25\pi}{2}</math></p> | <p>1 mark<br/>No penalty<br/>for not<br/>rationalising</p> <p>1 mark</p> <p>1 mark</p> | <p><b>Question 4</b></p> <p>i) <math>T = 24 - Ae^{-kt}</math><br/> <math>\frac{dT}{dt} = -k \times -Ae^{-kt}</math><br/> <math>\frac{dT}{dt} = -k(24 - Ae^{-kt} - 24)</math><br/> <math>\frac{dT}{dt} = -k(T - 24)</math><br/> <math>\therefore T = 24 - Ae^{-kt}</math> is a solution</p> <p>ii) when <math>t = 0, T = -10</math><br/> <math>-10 = 24 - Ae^0</math><br/> <math>\therefore A = 34</math></p> <p>iii) when <math>t = 90, T = 2^\circ C</math><br/> <math>2 = 24 - 34e^{-k \times 90}</math><br/> <math>\therefore \frac{-22}{-34} = e^{-90k}</math><br/> <math>\log_e \left(\frac{11}{17}\right) = -90k</math><br/> <math>k = -\frac{1}{90} \log_e \left(\frac{11}{17}\right)</math></p> <p>iv) when <math>t = 180 \text{ min}</math><br/> <math>T = 24 - 34e^{2 \log_e \left(\frac{11}{17}\right)} = 9.76</math></p> | <p>1 mark for<br/>complete<br/>answer</p> <p>1 mark</p> <p>1 mark</p> <p>1 mark</p> <p>1 mark</p> |  |

2022 Year 11 Mathematics Ext1 AT2 Solutions

**Question 5**

$$\begin{aligned} \text{ai) } LHS &= \frac{1}{\sqrt{x}} - \frac{1}{\sqrt{x+h}} \\ &= \frac{\sqrt{x+h} - \sqrt{x}}{\sqrt{x}\sqrt{x+h}} \times \frac{\sqrt{x} + \sqrt{x+h}}{\sqrt{x} + \sqrt{x+h}} \\ &= \frac{x+h-x}{x\sqrt{x+h} + \sqrt{x}(x+h)} \\ &= \frac{h}{\sqrt{x}(x+h) + x\sqrt{x+h}} \end{aligned}$$

$$\begin{aligned} \text{ii) } \frac{d}{dx} \sqrt{\frac{2}{x}} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \sqrt{2} \lim_{h \rightarrow 0} \frac{1}{h} \left( \frac{1}{\sqrt{x+h}} - \frac{1}{\sqrt{x}} \right) \\ &= \sqrt{2} \lim_{h \rightarrow 0} \frac{1}{h} \left( \frac{-h}{\sqrt{x}(x+h) + x\sqrt{x+h}} \right) \\ &= \sqrt{2} \left( \frac{-1}{x\sqrt{x} + x\sqrt{x}} \right) \\ &= \frac{-\sqrt{2}}{2x\sqrt{x}} = \frac{-\sqrt{2x}}{2x^2} \end{aligned}$$

**Question 6**

$$\begin{aligned} \text{a) } \sin x + \cos x &= -1 \\ \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} &= -1 \\ \therefore 2t + 1 - t^2 &= -1 - t^2 \\ 2t &= -2 \\ t &= -1 \\ \therefore \frac{x}{2} &= \pi - \frac{\pi}{4} \\ x &= \frac{3\pi}{2} \end{aligned}$$

test:  $x = \pi$

$$\sin \pi + \cos \pi = 0 - 1 = -1$$

$$\therefore x = \pi \text{ and } x = \frac{3\pi}{2}$$

$$\text{b) } LHS = \tan 2\sigma (\sin 2\sigma - \tan \sigma)$$

$$\text{let } t = \tan \sigma$$

$$\therefore LHS = \frac{2t}{1-t^2} \left( \frac{2t}{1+t^2} - t \right)$$

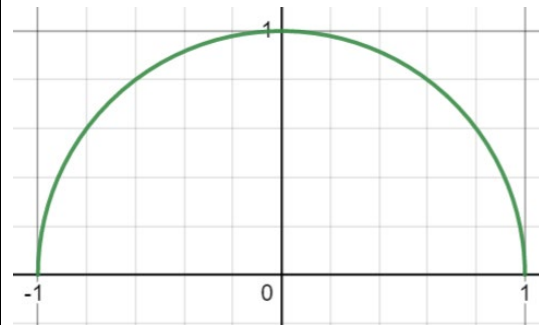
$$= \frac{2t}{1-t^2} \left( \frac{2t-t-t^3}{1+t^2} \right)$$

$$= \frac{2t}{1-t^2} \times \frac{t(1-t^2)}{1+t^2}$$

$$= \frac{2t}{1+t^2} \times t$$

$$= \tan \sigma \sin 2\sigma$$

c)



1 mark for intercepts

1 mark for the correct shape